Alicia de Cubas Yarza

aliciadecubas@gmail.com | *https://www.linkedin.com/in/alicia-de-cubas-yarza* | *https://aliciadecubas.github.io/*

ABOUT ME

I am a final-year Computer Science for Games student at Sheffield Hallam University, with a background in videogame programming. I enjoy creating responsive gameplay systems and intuitive player interfaces, as well as exploring the engine and graphics technologies behind games. Through projects using Unreal Engine, Unity, C++, OpenGL and Vulkan, I have developed experience in gameplay programming, rendering and technical problem-solving. I enjoy collaborating with others and turning ideas into playable experiences.

TECHNICAL SKILLS

- **Languages**: C++, C#, C, Python, Lua, Kotlin, GLSL, HLSL
- **Engines**: Unreal Engine and Unity, covering gameplay systems, UI, audio and interactive mechanics
- **Graphics and APIs**: OpenGL, Vulkan, shaders, lighting, materials, particle systems, procedural generation and terrain
- **Libraries**: Dear ImGui, GLM, GLFW, SDL, stb_image, miniaudio, OpenAL, nlohmann/json
- **Console development**: Developing a custom game engine in C++ for PS5 on PlayStation 5 Development Kits, rendering, memory management
- **Tools and version control**: Git, GitHub, debugging, profiling
- **Platforms**: Windows, Linux, PS5

EDUCATION

- **Computer Science for Games** (2026-2027)
  BSc (Honours)
  Sheffield Hallam University

- **HND in Computing (Game Programming)** (2023-2026)
  HND Degree
  ESAT

PROJECTS

- **NAOS01: The Quiet Sector | Unreal Engine** (2026)
  - Developed as a final university project: a stealth game focused on tension, exploration and responsive gameplay systems.
  - Implemented the HUD, player interaction systems, radar/sonar and audio integration.
  - Developed NPC behaviour to support the game's stealth gameplay.

- **Graphics Engine | C++ / OpenGL** (2025)
  - Contributed to a team-built 3D graphics engine at ESAT.

- Worked on procedural terrain, GPU buffers, materials, textures, particles, geometry and lighting.
- Contributed to the engine's rendering systems and visual features.

▪ **Procedural Terrain | Unity** (2025/2026)
- Developed a Unity project focused on procedural terrain and real-time environment generation.
- Built terrain-generation and world-building functionality for creating dynamic environments.

▪ **Bullet Hell Game | Unity** (2025)
- Developed a bullet hell game as part of a team project.
- Implemented the player controller, enemy firing patterns, HUD and parallax background movement.
- Contributed to the core combat and gameplay experience.

▪ **Weapon & Inventory Demo | Unreal Engine** (2025)
- Built the player weapon system and inventory flow for a team-based shooter prototype.
- Implemented weapon management and progression logic.
- Developed gameplay systems supporting the player's combat experience.

▪ **OpenGL/ Vulkan Graphics Engine | C++ / OpenGL / Vulkan** (2025)
- Developed a graphics engine in a team of three, supporting both OpenGL and Vulkan.
- Contributed foundational OpenGL components and implemented the Vulkan backend.
- Built a compatibility layer connecting the two graphics APIs.

▪ **Voice-Controlled Stealth Game | Unity** (2025)
- Created a two-person Unity game exploring microphone-based movement as an alternative to keyboard input.
- Implemented voice-controlled movement, with speaking volume affecting the player's speed.
- Designed the stealth objective around reaching the kitchen before time runs out without alerting the player's parents.

LANGUAGES

▪ Spanish (Native)
▪ English (C1 - IELTS Band Score 7.0 / 8.0)
▪ Valenciano (C1)